

Ujaan Mukherjee

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B.Tech CSE student, building agentic AI systems, RAG pipelines, and automation (LangChain, n8n, Docker). Comfortable across the stack, from low-latency backend systems to embedded/IoT, with a growing focus on low-level programming and Linux internals, working toward GSoC and kernel contributions.

EXPERIENCE

Intern | Agenscience.ai

June 2026 - August 2026

- Designed and built 3 AI-powered prototypes for Agenscience's decision-intelligence platform and client engagements
- Ran the full ML lifecycle, data exploration, feature engineering, model development, and evaluation, on client datasets
- Rapidly prototyped 2+ internal tools and proof-of-concept demos, accelerating team iteration cycles
- Designed and tested multi-agent LLM pipelines with tool-use and agentic orchestration, presented across 5+ team learning sessions

Technical Director | JAAGO NGO, Kolkata

2026 - Present

- Own all technical operations for the NGO, running its website and core systems end to end
- Built and shipped the organization's website solo, eliminating all third-party vendor costs

Physics Virtual Labs Intern | IEM IIFR, Kolkata

May 2026 - August 2026

- Develop virtual physics lab simulations used for hands-on student instruction at IEM-IIFR
- Validated and debugged simulation models, improving physical accuracy by ~15%

Secretary, Space Byte Club | Don Bosco School, Liluah

April 2024 - March 2025

- Ran end-to-end club operations as elected Secretary
- Organized and hosted 10+ technical events reaching 100+ students

PROJECTS

Argus: AI Semantic Face Search System

github.com/KireinaR/argus

Python, CLIP, NumPy, GPU

- Engineered a CLIP-based semantic face search system that cut query time by ~70%
- Optimized the embedding pipeline across 14,000+ images, slashing disk I/O by ~60%

Heat Seeker: Real-Time Thermal Tracking System

github.com/KireinaR/heat-seeker

ESP32, MLX90640, Python, OpenCV

- Engineered an ESP32 + MLX90640 thermal tracker achieving sub-100ms end-to-end latency with Python and OpenCV
- Delivered heat-signature detection and tracking at 90%+ accuracy
- Designed fully self-contained, plug-and-play deployment in under 5 minutes with zero cloud dependency

Global Air Bases REST API

airbases.mris.workers.dev

Hono, Cloudflare Workers, REST

- Built and deployed a global REST API covering 2,000+ air bases across 50+ countries on Cloudflare Workers, serving responses in under 10ms
- Architected a scalable geospatial schema with query filtering, running at zero server cost

YDSE Simulator: Interactive Physics Web App

ydse-ujaan.vercel.app

JavaScript, Physics Modelling, Vercel

- Built a physics-accurate double-slit experiment simulator sustaining 60 FPS across all major browsers
- Implemented wave interference and diffraction modelling from first principles

JAAGO NGO Website

jaagoforchange.vercel.app

HTML, CSS, JavaScript, Vercel

- Led the development team behind the NGO's public website
- Built the core frontend components and drove the UX direction
- Shipped a responsive, performance-optimized build on Vercel

TECHNICAL SKILLS

Languages	Python, JavaScript, Java, C++, Bash
AI & Machine Learning	CLIP, RAG, LangChain, LLM Fine-Tuning, Prompt Engineering, Transformer Models, Generative AI, NLP
Automation	n8n, AI-augmented development, intelligent tooling integration
Backend & APIs	Node.js, Express.js, Hono, Cloudflare Workers, REST API design
DevOps & Infrastructure	Docker, Git, Vercel, WSL, Oracle VirtualBox
Embedded Systems	ESP32, ESP8266, Arduino, OpenCV, real-time data streaming
Frontend	HTML, CSS, JavaScript

EDUCATION

Institute of Engineering and Management (IEM), Salt Lake, Kolkata	2025 - Present
B.Tech, Computer Science and Engineering	
Don Bosco School, Liluah, CISCE Board	2024 - 2025
Class 12, Science Stream: 89.5%	
Don Bosco School, Liluah, CISCE Board	2022 - 2023
Class 10: 93%	

CERTIFICATIONS

IBM	Fundamentals of AI Agents Using RAG and LangChain; Generative AI Advanced Fine-Tuning for LLMs; Generative AI Engineering and Fine-Tuning Transformers; Generative AI Foundational Models for NLP and Language Understanding; Generative AI: Introduction and Applications; Generative AI: Prompt Engineering Basics; Introduction to Artificial Intelligence
Google / Google Cloud	Introduction to Generative AI; Accelerate Your Job Search with AI; Use AI Responsibly; Stay Ahead of the AI Curve
Duke University	GenAI Basics: How LLMs Work; Python Scripting
Coursera	Perplexity AI for Beginners: AI-Powered Search